AM811001 Analytics for Managers

Using Data Analysis for Decision Making in Agriculture.

Assignment 1

Introduction:

Data-analytics thinking pertains to the skill of utilising data to guide decision-making and address intricate business issues (Ascarza, 2018). According to Wang (2017), it is approach requires managers to have a deep understanding of the data and analytics tools at their disposal and the ability to analyse and interpret data to draw insights and make informed decisions. The key benefit of data-analytic thinking is the ability to make data-driven decisions based on objective evidence rather than intuition or personal experience. This approach can lead to more accurate and reliable decision-making, as well as improved efficiency and cost savings (Brynjolfsson et al., 2011).

However, data-analytic thinking also presents several challenges for managers. First, it requires significant investment in data analytics tools and resources, which may be costly and time-consuming. Second, it requires a deep understanding of data analysis techniques, which may not be within the skillset of all managers. Finally, it can be difficult to balance data-driven decision-making with other factors, such as organisational culture and stakeholder expectations (Nielsen & Nielsen, 2020).

To overcome these challenges, managers should prioritise investing in data analytics tools and resources and providing employees with the necessary training and support to develop their data analysis skills. Additionally, managers should seek to strike a balance between data-driven decision-making and other factors, such as organisational culture and stakeholder expectations (Cederstav et al., 2020).

Data analytics thinking and data-driven decision-making are crucial in addressing various agricultural challenges. Agricultural logistics, precision farming, and pest and disease management are among the key areas where data analytics has been successfully applied to improve decision-making. This report discusses the context, problems, decisions, data, and justification for data analysis in these agricultural areas.

1. **Pest and disease management**

Pest and disease management is one of the most challenging aspects of agriculture. Effective pest and disease management requires accurate and timely information about crops' pest and disease threats. Data analysis can provide insights into the patterns and trends of outbreaks, enabling managers to make informed decisions about prevention, treatment, and control (Madasamy et al., 2020).

As per Ribarics (2016), the main challenge in pest and disease management is identifying and responding to outbreaks promptly and effectively. The decision to apply pesticides or other treatments must be based on accurate information about the pest or disease, the stage of the infestation, and the potential risks and benefits of different treatment options (Tseng et al., 2019).

By analysing data on pest and disease incidence, weather patterns, crop growth, and other factors, managers can identify farm or field areas at risk of infestation and take preventive measures to reduce the risk of outbreaks. They can also use data analysis to evaluate the effectiveness of different treatment options and optimise their use of pesticides and other treatments (Ribarics, 2016).

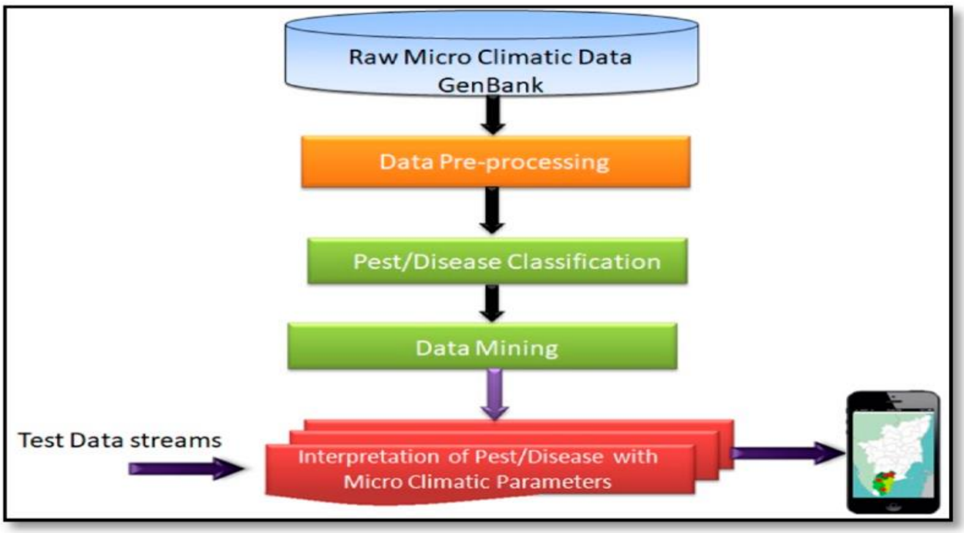

Figure 1- Using Data GenBank for data analysis

The data can come from various sources, including weather stations, crop sensors, satellite imagery, and pest monitoring systems (Figure 1). This data can be combined with historical data on outbreaks and other factors to create predictive models that can help managers to anticipate and respond to pest and disease threats (Madasamy et al., 2020).

The complexity and variability of the problem justifies the use of data analysis. Various factors, including weather patterns, soil conditions, crop management practices, and other pests and diseases, can influence pest and disease outbreaks. Data analysis can help managers to identify the most important factors and develop effective strategies for preventing and controlling outbreaks. It can also help them to evaluate the effectiveness of different treatments and adjust their management practices over time (Balan et al., 2020). Martin-Clouaire (2017) proposed predictive modelling for pest outbreaks: by analysing historical data on outbreaks, weather patterns, and crop growth, managers can develop predictive models that can help them to anticipate and respond to future outbreaks. For example, a study conducted by researchers at the University of Nebraska-Lincoln used a sensor network to monitor the presence and abundance of soybean aphids in real time. This information was used to guide decisions about when and where to apply insecticides, resulting in a reduction in insecticide use and increased crop yields (Jhala et al., 2021). Madasamy (2020) compares actual data with predicted data and uses analytics to assess the accuracy of the forecast (Figure 2).

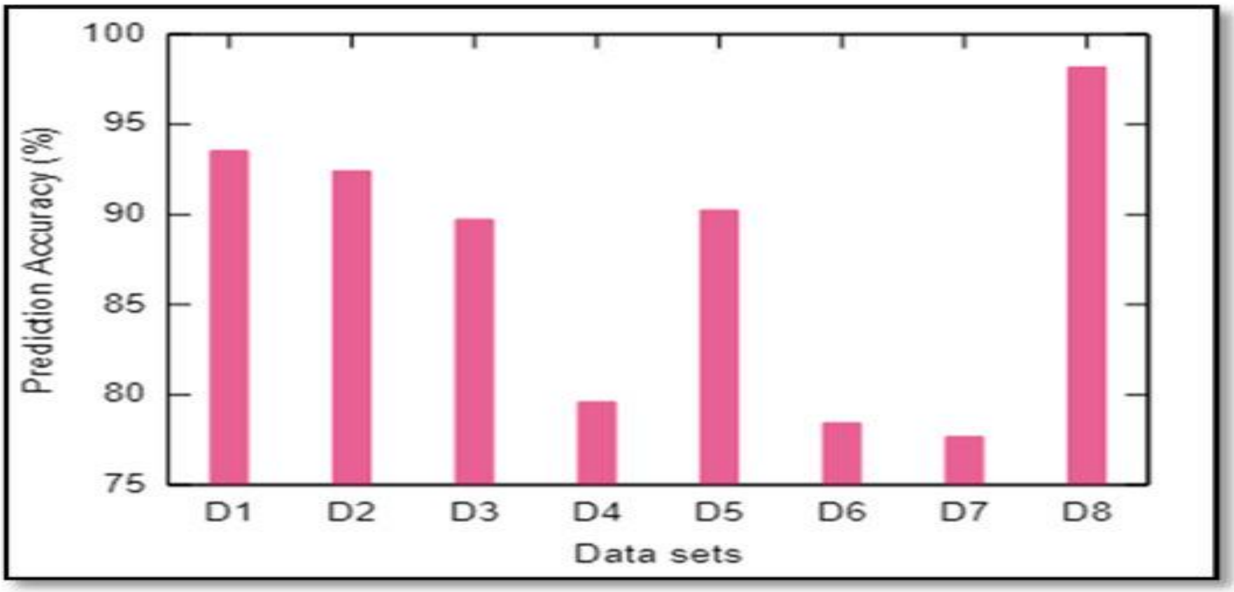

Figure 2- Prediction accuracy of wheat yield.

Data analysis is a powerful tool for decision-making in pest and disease management, providing insights into the patterns and trends of outbreaks and guiding decisions about prevention, treatment, and control. Using data analysis to develop predictive models, monitor pest populations in real-time, and assess disease risk, managers can make informed decisions that reduce the risk of crop damage and increase the sustainability of agricultural production. However, managers need to ensure that the data used in their analysis is accurate and reliable and that they have the necessary expertise to interpret the results effectively (Bronson & Knezevic, 2016).

2. Precision agriculture

Precision agriculture uses data and technology to manage agricultural production more efficiently and sustainably. Data analysis plays a critical role in decision-making in precision agriculture, providing insights into soil health, crop growth, and environmental conditions that can help managers to optimise inputs, reduce waste, and increase yields. In this report, we will explore how data analysis helps in decision-making in precision agriculture, focusing on three key areas: soil management, crop management, and yield optimisation. However, precision agriculture also presents challenges, including the need for accurate and reliable data and managers to interpret and act on that data effectively (Griffin et al., 2017).

One of the key challenges in precision agriculture is managing soil health. Soil health is essential to crop growth but cannot be easy to measure and manage effectively. Soil analysis provides critical data on nutrient levels, pH, and other factors influencing soil health, but this data can be complex and difficult to interpret (Bhat & Huang, 2021). Another issue in precision agriculture is managing crop growth and development. Monitoring crop growth and development is essential to optimising inputs and yields, but it requires accurate and timely data on environmental conditions, soil health, and other factors that influence crop growth. Finally, yield optimisation is a key problem in precision agriculture. Yield optimisation involves optimising inputs and crop management practices to maximise yields while minimising waste, but it requires accurate and reliable data on crop growth, environmental conditions, and other factors that influence yield (Sagar & N K, 2018). Using special sensors and IOT, managers can collect accurate and relevant data for analysis (Figure 3).

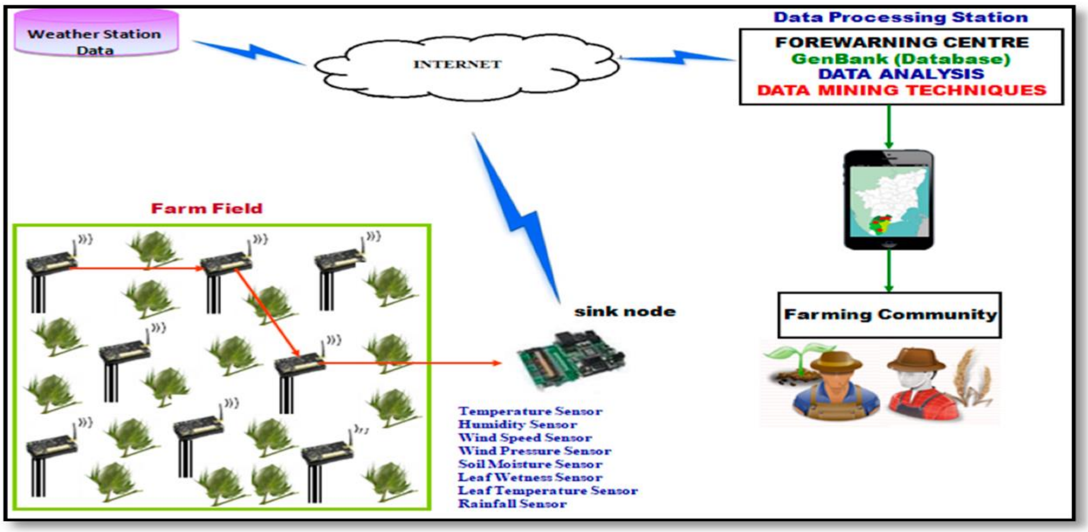

Figure 3- Collecting and using data in precision agriculture

Data analysis can help managers make informed decisions in these areas. In soil management, data analysis can provide insights into soil health, helping managers to improve inputs and reduce the risk of nutrient depletion or other soil health problems. In crop management, data analysis can provide real-time insights into crop growth and development, helping managers to optimise inputs and respond quickly to changes in environmental conditions or other factors that influence crop growth. In yield optimisation, data analysis can help managers to identify areas where inputs can be optimised, reducing waste and increasing yields (Griffin et al., 2017).

Data used in precision agriculture can come from a variety of sources, including soil analysis, sensor networks, weather stations, and remote sensing technologies. Sensor networks and weather stations provide real-time data on environmental conditions, including temperature, humidity, and precipitation. Remote sensing technologies, such as satellite imagery, can provide insights into crop growth and development, including plant health, leaf area index, and chlorophyll content (Mintert et al., 2016).

The use of data analysis in precision agriculture is justified by the potential economic and environmental benefits of optimising agricultural production. By optimising inputs and reducing waste, precision agriculture can help to reduce costs and increase yields while minimising the environmental footprint of agricultural production (Bronson & Knezevic, 2016). With careful planning and analysis, data-driven decision-making can support more sustainable agriculture.

For example, in precision agriculture, data can be used to determine the ideal seeding rate for a particular crop or compare the correlation between weather conditions and yield (Figure 4). By analysing soil data, weather patterns, and other factors, farmers can determine the optimal number of seeds to plant per acre, reducing waste and increasing yield (Ribarics, 2016).

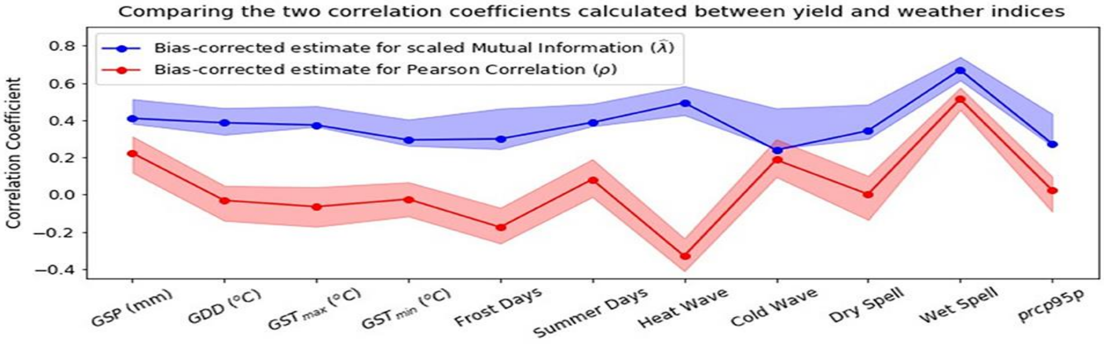

Figure 4 - comparing the correlation between weather conditions and yield

3. Agricultural logistics

Agricultural logistics is the process of managing the flow of agricultural products, information, and other resources between the point of origin and the point of consumption. This process involves a range of activities, including transportation, storage, packaging, and distribution. Data analysis plays a crucial role in agricultural logistics decision-making (Dupal' et al., 2019). As per Cao (2022), weather conditions, transportation costs, storage facilities, and market demand often impact logistics.

The key challenges faced in agricultural logistics include:

-Inefficient transportation and storage: the lack of proper transportation infrastructure and storage facilities can lead to product spoilage, delays, and increased transportation costs.

-Unpredictable demand: the fluctuation in market demand and the inability to predict demand accurately can result in underutilisation or overstocking inventory.

-Traceability: it is crucial to maintain a traceable system in agricultural logistics, from farm to the final point of consumption, to ensure the safety and quality of the products (Dejun & Guangsheng, 2020).

Various types of data in agricultural logistics are significant to making informed decisions. Figure 6 presents the data flow. As per Dejun & Guangsheng (2020), these include:

-Transportation data: information on the mode of transport, transportation costs, transit time, and delivery performance.

-Inventory data: inventory levels, turnover, and demand forecasting data.

-Weather data: data on weather conditions, such as temperature, humidity, and precipitation, which impact transportation, storage, and shelf life of products.

-Traceability data: data on the origin, quality, and safety of products, which can help in managing recalls and preventing foodborne illness (Figure 5).

Variables	Estimate	Standard Error	Wald Chi-Square	Pr>Chi Square
Intercept	**1.35**	**.285**	**22.55**	**<.0001**
Land Area	**.011**	**.0024**	**20.76**	**<.0001**
Dist_Nonag	**.002**	**.0006**	**12.74**	**<.0001**
Dist_Inter State	**.00006**	**.00002**	**7.30**	**.0069**
Dist_Town	-.00001	.00001	.76	.3811
Dist_Fred	.00004	.00001	1.53	.21
NonIrryiel	**.065**	**.020**	**10.64**	**.0011**
PFA	**-.55**	**.217**	**6.49**	**.010**

N = 4504

Figure 5 – Estimation of logistics directions

Figure 6 – Data flow in logistics.

Data analysis plays a critical role in agricultural logistics, as it helps to identify areas of inefficiency and to optimise the logistics process. The use of data in agricultural logistics can help to:

a. Improve transportation and storage efficiency. Analysing transportation data makes it possible to identify the most cost-effective mode of transport, optimise routes, and reduce transit times. Moreover, analysing inventory data helps to optimise inventory levels and reduce waste.
b. Predict market demand. By analysing demand data, it is possible to predict market demand accurately and improve production and inventory accordingly.
c. Ensure traceability and food safety: By analysing traceability data, it is possible to track products from the farm to the final point of consumption, ensuring product safety and quality.

Agricultural optimisation can become more productive and sustainable by using data to optimise logistics, improve efficiency, reduce costs, and enhance customer satisfaction. The use of data-driven decision-making in agricultural logistics is essential to meet the challenges faced in the agricultural industry and to ensure a smooth and efficient flow of products from the farm to the final point of consumption (Brynjolfsson et al., 2011).

4. Conclusion.

Data analytics thinking and data-driven decision-making are crucial in addressing various agricultural challenges. Agricultural logistics, precision farming, and pest and disease management are among the key areas where data analytics has been successfully applied to improve decision-making. In agricultural logistics, data analytics helps to optimise supply chain management and reduce inefficiencies. Precision farming enables farmers to make more accurate decisions regarding crop management, thereby improving yields and reducing costs. In pest and disease management, data analytics provides insights that help farmers to detect and prevent outbreaks, reducing the risk of crop damage and loss (Wang, 2017).

The successful implementation of data analytics in agriculture requires access to accurate and reliable data and advanced analytical tools and techniques. As such, investments in technology, data collection, and analysis capabilities are crucial to enable effective agricultural decision-making (Kruse et al., 2018).

However, there are still challenges to overcome, including limited access to data, insufficient analytical skills among farmers, and the high cost of implementing data analytics solutions. Governments, private sector companies, and other stakeholders must work together to address these challenges and promote the adoption of data analytics in agriculture (Mintert et al., 2016).

Data analytics thinking and data-driven decision-making have enormous potential to transform the agricultural industry. By enabling farmers to make better decisions and improve their operations, data analytics can contribute to increased productivity, improved sustainability, and enhanced profitability in agriculture (Bhat & Huang, 2021).